EXPERIMENT 1

Computer Networks — Basic Network Commands

Aim: To execute and study basic network commands such as ipconfig, netstat, nslookup, arp and hostname.

1. Command: `ipconfig /all`

Displays the full IP configuration of the Windows host: host name, node type, adapter details, physical (MAC) address and DHCP status.

```
C:\Users\DELL>IPCONFIG /ALL

Windows IP Configuration

   Host Name . . . . . . . . . . . . : HDC0124111
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Ethernet Connection (17) I219-LM
   Physical Address. . . . . . . . . : 20-88-10-86-7A-AC
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

Fig 1: Output of ipconfig /all

2. Command: `netstat`

Lists active Internet connections and UNIX domain sockets with protocol, state, I-Node and path.

```
student@fedora:~127$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 fedora:bootpc           _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     22172
unix  2      [ ]         DGRAM                    19789    /run/systemd/home/notify
unix  3      [ ]         STREAM     CONNECTED     23322
unix  3      [ ]         STREAM     CONNECTED     20679    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     23105    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     22318    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     18295
unix  3      [ ]         STREAM     CONNECTED     17619
unix  3      [ ]         STREAM     CONNECTED     24004
unix  3      [ ]         STREAM     CONNECTED     22966    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     23195
unix  2      [ ]         DGRAM      CONNECTED     18381    /run/chrony/chronyd.sock
unix  3      [ ]         STREAM     CONNECTED     22814    /run/systemd/journal/stdout
unix  2      [ ]         DGRAM                    20131
unix  3      [ ]         STREAM     CONNECTED     19881    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     22491    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     23101    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     22869    /run/systemd/journal/stdout
unix  2      [ ]         DGRAM      CONNECTED     21463
unix  3      [ ]         STREAM     CONNECTED     23265
unix  3      [ ]         STREAM     CONNECTED     22246
unix  3      [ ]         STREAM     CONNECTED     22196    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     21406    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     19708    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     23394
unix  3      [ ]         STREAM     CONNECTED     23342    /run/user/1000/pipewire-0
unix  3      [ ]         STREAM     CONNECTED     23572
```

Fig 2: Output of netstat

3. Command: `nslookup`

Queries the DNS server to resolve the domain name www.google.com to its IP addresses.

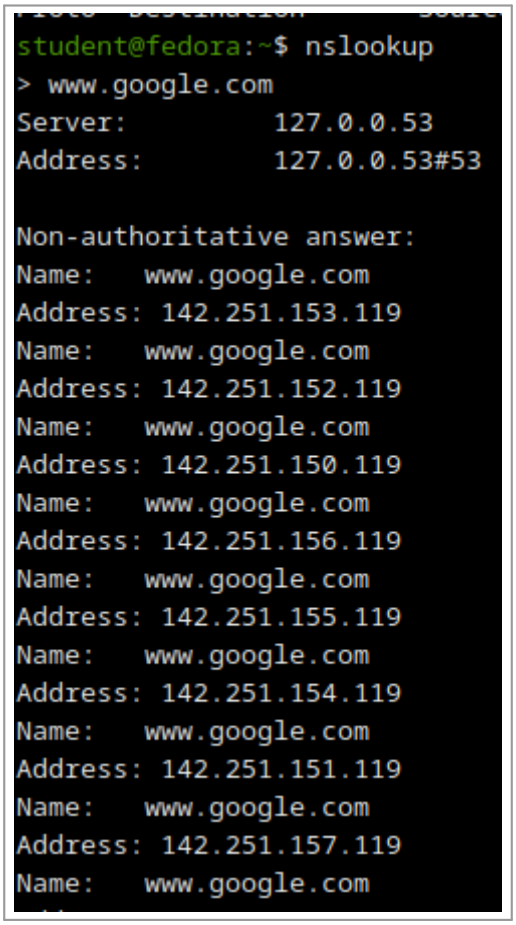


Fig 3: Output of nslookup

4. Command: `arp -a & hostname`

arp -a shows the ARP cache (IP to MAC mapping of the gateway); hostname prints the name of the current machine.

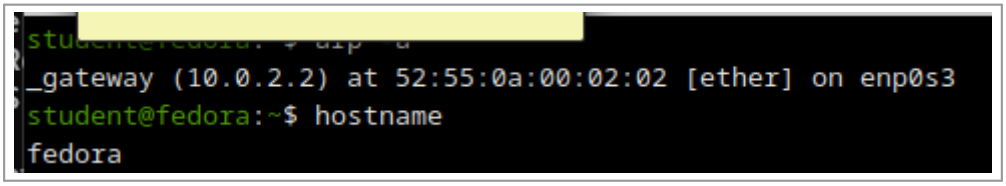


Fig 4: Output of arp -a & hostname

Result

The basic network commands were executed successfully and their outputs were observed and verified.